Investigation – how does the length of a bone affect its strength?

Aim

To find out how the strength of a tube of paper (bone) changes as its length increases. I will do this by hanging weights on the 'bone' until it breaks.

Independent and dependent variables

A variable is something in your experiment that changes. It may change because *you* change it (this is the **independent variable**) or it could change as a result of doing the experiment and you are going to measure this change (the **dependent variable**).

Tasks

A. For this investigation, write down the independent and the dependent variable.

- independent variable – the variable you are going to control.
- dependent variable – the variable that you will measure when you do the experiment.

B. Draw a table to input the following data (choose the appropriate units for each entry);

- length of bone (there are 5 lengths to investigate)
- diameter of bone
- mass needed to break bone.

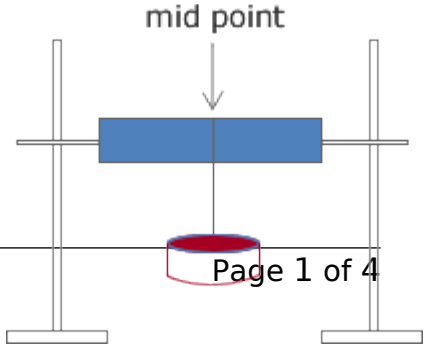
C. You do not want any other variables to change. These are ***control variables*** and must be kept the same. Read through the method below and write down the control variables.

D. Make five paper bones using the following instructions. The bones should be of different lengths – 20cm, 30cm, 40cm, 50cm and 60cm.

1. Take the longest paper (60cm) and roll it up tightly around a pencil.
2. Use a 5cm length of sellotape to hold the curled paper in place in the middle. Remove the pencil.
3. Measure the diameter of the bone accurately.
4. Mark the mid-point of the 'bone'.
5. Repeat this for each of the bones.
6. Remember they should all be made using paper of the same width and rolled to the same diameter.
7. We will assume the thickness of each piece of paper is the same.

E. Test the strength of the bones using the following method.

1. Set up the stands as shown in the diagram.
2. Rest the longest (60cm) bone on the clamps.



3. Hang a 100g weight on the midpoint of the bone.
4. Keep adding weights until the ‘bone’ breaks (bends).
5. Record the data in your table.
6. Repeat steps 5-8 with the other 4 lengths of bone.
7. Draw a **graph** of your results and make a **conclusion**.

Teaching notes

Context

Students investigate how the strength of a tube of paper (bone) changes as its length increases. They do this by hanging weights on the mid-point of the 'bone' until it breaks (bends). This investigation could feature whilst teaching a topic on bones, joints or movement, but it is meant to build on their practical skills. It requires students to do the following;

- *identify an independent and dependent variable*
- *pick out control variables and understand why this is important*
- *draw a table for their results*
- *carry out an investigation in a methodical way and record accurate data*
- *draw a graph of the data with a line of best fit*
- *make a conclusion.*

ISAs at GCSE level require students to understand and use *variables*.
The teacher can discuss the different types of variables with the students – *independent*, *dependent* and *control* variables.

In this investigation the variables are;

- independent – length of bone
- dependent – mass required to break the bone
- control – diameter of bone, width of paper, length of sellotape, weights put onto midpoint of bone, paper thickness, overhang of bone on clamp.

Emphasis should be placed on the need to make the test *valid*. If more than one independent variable were to change, the experiment would not be valid.

Equipment per group

- 2 x retort stands, clamps and bosses.
- 100g weights with holder and hook (wide enough to hang over the bone)
- 5 x 5cm lengths of sellotape
- ruler
- 5 sheets of paper (one each of the following dimensions): 20cm x 21cm
  30cm x 21cm
  40cm x 21cm
  50cm x 21cm
  60cm x 21cm.

The paper used in the experiment should be cut to size before the lesson and preferably marked with an arrow to indicate the way it should be rolled up.

The investigation can get a bit noisy with falling weights so encourage your students to put their hands under the weights to catch them.